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To cite this version:

Sarah Theroine, Laurent Gautier, Christophe Cruz, Gennaro Ascione. Textual Molds for In-Context Learning at Sea. Wenn KI auf Fach und Sprache trifft, 12, Frank & Timme, pp.185-202, 2026, Studien zu Fach, Sprache und Kultur, 978-3-7329-8690-3. ⟨10.57088/978-3-7329-8690-3_10⟩. ⟨hal-05647751⟩

HAL Id: hal-05647751

https://hal.science/hal-05647751v1

Submitted on 3 Sep 2026

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Textual Molds for In-Context Learning at Sea

Can Generative AI Produce Shipping Forecasts without Semantic Guidance?

Abstract

Seewetterberichte stellen ein stark eingeschränktes Genre der Fachkommunikation dar. Sie sind auf operative Effizienz und Sicherheit ausgelegt und unterliegen strengen Normen. Sie zeichnen sich durch eine kontrollierte Sprache, feste lexikalisch-grammatikalische Strukturen und semantische Vorhersehbarkeit aus und weisen das auf, was Sinclair (1991) als Textmuster und Condamines (2018) als Diskursfixiertheit bezeichnen. Der Beitrag untersucht, ob LLMs diese strukturellen und konzeptuellen Einschränkungen replizieren können. Es wurde zuerst eine Korpusanalyse französischer Vorhersagen durchgeführt, um wiederkehrende lexikalische und syntaktische Muster zu identifizieren, die das Genre definieren. Sie wurden dann mit den zugrunde liegenden semantischen Strukturen artikuliert. Diese Muster dienen schließlich der Bewertung der Ergebnisse von zwei LLMs (Claude-opus-4 und Grok-4-fast), die unter Zero-Shot-Bedingungen generiert wurden. Die Ergebnisse zeigen, dass beide Modelle zwar oberflächliche Regelmäßigkeiten erfassen, jedoch häufig von den domänenspezifischen

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konzeptuellen Strukturen abweichen. Dies wirft weitergehende Fragen hinsichtlich der Genre-Sensitivität, der epistemischen Kohärenz und der Grenzen der promptbasierten Textgenerierung für spezialisierte Kommunikationsaufgaben auf.

Keywords AI, linguistic patterns, LSP, shipping forecast, textual mold

1 Introduction

Shipping forecasts represent one of the most highly codified genres within professional communication, characterized by standardized phraseology, condensed syntax, and predictable surface structures. As a type of controlled language, their narrow lexical range and grammar ensure clarity, safety, and international intelligibility in high-stakes contexts (Bocanegra-Valle 2011, 2015; Condamines 2018). This study focuses on French shipping forecasts issued by Météo-France. These can be considered a subtype of specialized discourse shaped by institutional and international regulations, including the IMO[2] and WMO[3] standards governing the Global Maritime Distress and Safety System (GMDSS)[4].

This study is guided by the following research question: To what extent can LLMs reproduce both the surface patterns and the underlying conceptual constraints of shipping forecasts? Due to their formulaic nature and semantic regularity (Gautier 2008), these texts offer a robust case study to evaluate the genre sensitivity of Large Language Models (LLMs). This paper explores whether two state-of-the-art models — Claude-opus-4 (dense architecture) and Grok-4-fast (Mixture-of-Experts) — can replicate both the surface and conceptual structures of shipping forecasts in zero-shot conditions. Our method therefore combines corpus linguistics, frame semantics and computational

2 International Maritime Organization (IMO): https://www.imo.org/

3 World Meteorological Organization: https://wmo.int/about-us/governance/technical-commissions/standards-and-recommended-practices

4 Global Maritime Distress and Safety System: https://gmdss.com/

text generation. A manually annotated corpus of Météo-France forecasts was used to identify recurrent patterns and core frame elements (FEs). The LLMs were then prompted to generate French forecasts, allowing us to assess their ability to reproduce constrained, domain-specific discourse without explicit semantic input.

This article is structured as follows: Section 2 briefly reviews relevant literature, section 3 outlines the methodology applied, section 4 presents the experimental setup, section 5 reports findings, and section 6 discusses implications and future directions.

2 State-of-the-art

Shipping forecasts are structured sequences of fixed lexico-grammatical patterns, each serving distinct semantic and discursive functions. Bocanegra-Valle (2015) describes them as "straightforward and highly conventionalized texts", thus that such texts are patterned, standardized and highly conventional, a perspective that can be further refined by emphasizing their sequential organization. Rooted in operational routines that prioritize clarity and safety, these texts exemplify controlled language, characterized by restricted vocabulary, elliptical syntax, and minimal variation (Condamines 2018; Johnson 1995) all the while benefitting from scientific linguistic knowhow. The concept of usability, central in ergonomics as it puts people at the heart of needs, is adapted to the perspective of ergonomic linguistics. In order to put this branch of linguistics to work, we need to take into consideration on the one hand the real life corpora or language use in context, and on the other hand implement tests to evaluate their proper usage and relevance. We base our study on controlled languages because of the way they are constituted (the rules that make them). Despite their constraints, linguistic research has rarely addressed their patterned structure as organized templates. Sinclair's textual mold (1991) offers a productive lens for analyzing this genre, framing forecasts as epistemically structured templates that stabilize production and interpretation (Condamines 2018). Recent studies formalize these molds

through pattern extraction and semantic modeling, showing their link to domain-specific knowledge structures (Aussenac-Gilles/Condamines 2012; Orna-Montesinos 2012). As textual molds intertwine rigid phraseological patterns with domain-specific conceptual constraints, they present a particularly revealing challenge for generative models, which may imitate surface regularities without internalizing the underlying semantic architecture of the mold.

Recent advances in LLMs have renewed interest in replicating such constrained genres. Through in-context learning (ICL), models infer task structure from prompts alone, even in zero-shot conditions, mimicking analogy-based reasoning (Dong et al. 2024). This mechanism is suited to controlled language, allowing models to generalize from minimal structured input. Claude-Opus-4, a dense model optimized for long-context reasoning (200k tokens), emphasizes coherence and structured task performance. In contrast, Grok-4-fast adopts a Mixture-of-Experts (MoE) architecture, distributing input across specialized subnetworks to support faster and more flexible generation (256k tokens). Said models embody complementary approaches to constrained text generation: stable reasoning versus expert-routed contextual specialization.

3 Method

The first stage involved the manual extraction of recurring lexico-grammatical patterns from a corpus of French shipping forecasts. Its aim was to collect stable associations of lexical and syntactic units that characterize the genre. Special attention was devoted to morpho-syntactic constructions linking meteorological phenomena, measurements, movements, and spatio-temporal references, drawing on a constructionist perspective as proposed by Liégeois/Gautier (2025a, 2025b) and Liégeois et al. (2023). Said patterns reflect cognitively efficient strategies for encoding specialized knowledge in operational discourse and give major importance to semantic structures. Each pattern was segmented into conceptual elements, allowing for the abstraction of a conceptual sequence. Details on frame-semantic annotation procedures are

not presented here, as the method builds on Theroine et al. (2025). Although not used here for training, these references served to evaluate on an objective basis whether the outputs of LLMs were in line with the genre's formal and semantic regularities. The second stage employed ICL techniques to automate the generation of shipping forecast sentences. In a zero-shot setting, the models were prompted to generate new sentences, relying on the following instructions. This setting has been chosen because it forces the models to infer the genre's structural and semantic constraints from a single authentic instance, making it an effective test of true zero-shot generalization.

> "follow the example given to generate 10 new sentences on marine weather \nText:" Dépressions 1007 hPa au large d'Ouessant se creuse et se décale vers le sud-est prévue 1001 hPa sur les pays de la Loire ce soir. Elle se comble sur le centre de la France demain dimanche. + "\ngenerated sentence:"

4 Data description

The corpus consists of authentic shipping forecasts published by Météo-France. As specified in the introduction, such forecasts represent a recurrent and highly standardized form of expert communication, designed to ensure navigational safety and support decision-making at sea. Issued regularly for specific maritime zones, they follow an established discursive routine that promotes stability, precision, and interpretive predictability. Pritchard/Kalogjera (2000: 185) observe that these texts are characterized by "a limited vocabulary and simple grammatical structure to suit the specific requirements". The latter being: safety, reliability, and uniformity. Each forecast adheres to a fixed macrostructural sequence composed of successive informational sections (cf. Figure 1), or moves, corresponding to the communicative stages (Bocanegra-Valle 2015).

Origine Météo-France. Bulletin côtier pour la bande des 20 milles, de la baie de Somme au cap de la Hague, du jeudi 10 juillet 2025 à 18H15 légales.

Vent moyen selon échelle Beaufort. Mer selon échelle Douglas. Heure légale = heure UTC+1 en hiver et heure UTC+2 en été. Attention : en situation normale, les rafales peuvent être supérieures de 40 % au vent moyen et les vagues maximales atteindre 2 fois la hauteur significative.

Pas d'avis de vent fort en cours ni prévu.

Situation générale le jeudi 10 juillet 2025 à 12H00 UTC, et évolution

Dépression 1015 hPa à l'ouest du Portugal, se comblant vendredi après-midi. Anticyclone 1025 hPa au sud de l'Irlande avec dorsale sur la Manche, prévu 1023 hPa en Mer du Nord vendredi matin. Dépression se creusant 1013 hPa cette nuit dans le fond du Golfe de Gascogne, prévue 1008 hPa vendredi soir.

Observations le jeudi 10 juillet 2025 à 15H00 UTC

Cap de la Hague : vent Est-Nord-Est 6 nœuds, 1023 hPa en baisse, clair ou peu nuageux, visibilité 10 milles. Bernières : vent Nord-Nord-Est 6 nœuds. Cap de la Hève : vent Nord 6 nœuds, rafales 16 nœuds. Dieppe : vent Nord-Ouest 4 nœuds, 1023 hPa en baisse. Cayeux sur mer : vent Ouest 8 nœuds.

Figure 1: Macrostructure of marine weather forecast

The corpus compiled consists of written shipping forecasts collected between February and March 2024, covering the Western Mediterranean sea area.

Number of documents	Number of tokens	Number of types	Number of sentences
29	38,416	550	3,245

Figure 2: Quantitative information about the corpus

5 Results

5.1 Identifying standardized patterns and semantic frames

Shipping forecasts follow a constrained style, marked by elliptical syntax, nominal condensation, and frequent lexico-grammatical repetition. Verbs are rare (10 %) and mostly appear as participles (Theroine et al. 2025). Meaning is conveyed through dense noun and preposition phrases (cf. Figure 3) encoding space, time, and phenomena. This structure reflects the principles of controlled

languages (Condamines 2018) and Sinclair's (1991) notion of a textual mold shaped by institutional constraints.

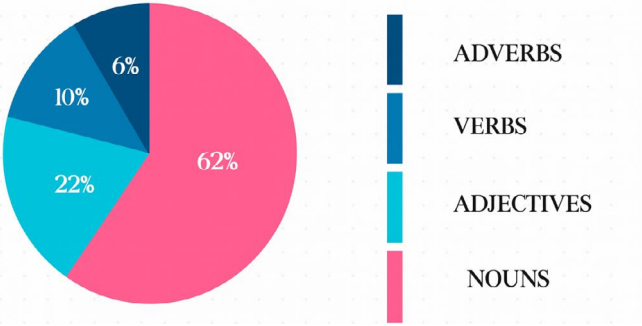


Figure 3: Distribution of grammatical classes in the corpus

Predicative verbs and their combinatory profiles (Gautier 2014) were extracted using Sketch Engine (Kilgarriff et al. 2004). Through targeted Corpus Query Language (CQL) searches, predicates were identified and mapped onto semantic frames following the FrameNet typology (Fillmore 2006). A methodological decision was to center CQL queries around the fixed expression *prévue* ("forecasted" or "expected"). This lexical item, although morphologically derived from *prévoir* ("to forecast" or "to anticipate"), has undergone semantic bleaching and grammaticalization: Rather than functioning as an inflected verb, *prévue* acts as a disciplinary figement, a stable discourse marker, signaling the transition from ongoing dynamics (*se creuser* – "to deepen", *se déplacer* – "to move") toward expected atmospheric states (*prévue 1001 hPa ce soir* – "forecasted 1001 hPa this evening").

French sentences	Translated versions
Dépressions 1007 hPa au large d'Ouessant se creuse et se décale vers le sud-est prévue 1001 hPa sur les pays de la Loire ce soir. Elle se comble sur le centre de la France demain dimanche	Low-pressure system at 1007 hPa off the coast of Ouessant is deepening and shifting southeastward, forecasted at 1001 hPa over the Pays de la Loire this evening. It will fill over central France tomorrow, Sunday.
Dépressions à 1007 hPa près des côtes bretonnes se creusant et se déplaçant vers le sud-est dans les prochaines heures, prévue 1001 hPa sur la région des Pays de la Loire ce soir. Elles s'affaiblissent progressivement sur la région lundi matin.	Low-pressure systems at 1007 hPa near the Brittany coast deepening and moving southeast over the next few hours, forecasted at 1001 hPa over the Pays de la Loire region this evening. They will gradually weaken over the region Monday morning
Tempêtes à 983 hPa au large de la Manche ce matin se déplaçant vers l'est dans les prochaines heures, prévue à 980 hPa sur la côte normande ce soir. Elles perdent en intensité sur le nord de la France demain matin.	Storms at 983 hPa off the English Channel this morning moving eastward over the next few hours, forecasted at 980 hPa over the Normandy coast this evening. They will weaken over northern France tomorrow morning.

Figure 4: Original weather sentences and their translated versions

In these sentences, *prévue* consistently appears in the same syntactic position and with the same semantic function, confirming its treatment as a non-generic but structurally essential slot, corresponding to the Expectation FE. It serves as a hinge between the description of the current atmospheric evolution and its projected state, thus encoding the prospective dimension crucial to forecast communication (Condamines 2018; L'Homme 2020). Due to its role, *prévue* serves as an effective anchor for extracting low-variability, formulaic sentences. Figure 4 consistently displays six core semantic slots: Phenomenon, Measure, Location, Movement, Time_period, and Expectation.

<phenomenon> <measure> <location> <time_period> <move> <expectation>

Therefore, based on the identified slots, we manually extracted the following lexico-grammatical patterns (cf. Figure 5).

First annotated sentence	Second annotated sentence	Third annotated sentence
Dépressions [phenomenon] 1007 [measure_value] hPa [measure_unit] au large d'Ouessant {location] se creuse [V. intensity] et se décale [V. Mouvement] vers le sud-est [Location] prévue [expectation] 1001 [measure_value] hPa [measure_unit] sur les pays de la Loire [Location] ce soir [time_period]. Elles [phenomenon] se comblent [V. movement] sur le centre de la France [location] demain dimanche [time_period].	Dépressions [Phenomenon] à 1007 [measure_value] hPa [measure_unit] près des côtes bretonnes [location] se creusant [V. intensity] et se déplaçant [V. Mouvement] vers le sud-est [location] dans les prochaines heures [time_period], prévue [expectation] 1001 hPa [measure] sur la région des Pays de la Loire [location] ce soir [time_period]. Elles [phenomenon] s'affaiblissent [V. intensity] progressivement sur la région [location] lundi matin [time_period].	Tempêtes [phenomenon] à 983 [measure_value] hPa [measure_unit] au large de la Manche [location] ce matin [time_period] se déplaçant [V. movement] vers l'est [location] dans les prochaines heures [time_period], prévue [expectation] à 980 hPa [measure] sur la côte normande [location] ce soir [time_period]. Elles [phenomenon] perdent en intensité [V. intensity] sur le nord de la France [location] demain matin [time_period].

Figure 5: Semantically annotated sentences

From these observations, a generalized semantic structure was derived:

> <phenomenon> <measure value> <measure unit> <location> <time_period> <move> <location> <time_period> <expectation> <measure> <location> <time_period>,

which is followed by a secondary structure describing the subsequent evolution:

> <phenomenon> <move> <location> <time_period>.

The structure does not merely describe a surface form, but rather captures a fully developed cognitive frame, rooted in the way forecasts mentally organize meteorological phenomena. Although the frame structure is generally stable, slight variation exists such as optional time markers and shifts between movement verbs (*se déplacer* – "to move") and intensity verbs (*se creuser* – "to deepen", *s'affaiblir* – "to weaken"), with intensity adding specificity to movement. Cognitively, such specifications are more relevant for sailors. Each lexical unit was aligned with its FEs, producing a pattern reflecting both linguistic regularity and conceptual coherence. This dataset served as a controlled-language reference to assess whether, under zero-shot conditions,

LLMs could replicate the structured, semantically constrained discourse under study, despite their limitations in generating epistemically grounded content (Bender et al. 2021).

5.2 Artificial shipping sentences

To assess how LLMs internalize the stylistic and structural regularities of shipping forecasts, both models were prompted to generate synthetic forecast sentences. The goal was to evaluate whether each model could autonomously reproduce the controlled-language features of the genre including precise scalar values, elliptical syntax, and event-driven sequencing.

Sentence No.	Claude-opus-4	Grok-4-fast
1	Dépression 1009 hPa au nord de la Bretagne se renforce et se déplace vers l'est prévue 1003 hPa sur la Manche centrale demain matin. Elle s'évacue vers la Mer du Nord lundi après-midi.	Tempête « Alex » 995 hPa au large de Brest se déplace vers l'est, prévue 990 hPa près des côtes vendéennes cette nuit. Elle s'affaiblit sur l'océan Atlantique demain lundi.
2	Anticyclone 1025 hPa sur l'Atlantique s'étend vers le golfe Gascogne prévu 1022 hPa près des côtes landaises ce soir. Il se maintient sur le sud-ouest mardi.	Front froid 1012 hPa au nord de la Biscaye s'intensifie et progresse vers le sud, atteignant 1008 hPa au large de Bilbao au lever du jour. Il se dissipe sur l'Espagne intérieure mardi après-midi.
3	Dépression 1004 hPa en Méditerranée occidentale se creuse rapidement prévue 999 hPa au sud de la Corse demain soir. Elle remonte vers le Golfe de Gènes mercredi matin.	Dépression méditerranéenne 1003 hPa près de Gênes se creuse rapidement vers l'ouest, prévue 999 hPa sur la côte ligure ce soir. Elle se comble au-dessus des Alpes demain matin.
4	Thalweg 1010 hPa sur les Baléares s'approfondit et se décale vers le nord-est prévu 1006 hPa près de la Sardaigne cette nuit. Il se comble progressivement jeudi.	Anticyclone atlantique 1030 hPa au large des Açores se renforce et s'étend vers le nord-est, prévu 1035 hPa sur les îles Britanniques vendredi soir. Il persiste sur l'Europe occidentale samedi.

Sentence No.	Claude-opus-4	Grok-4-fast
5	Dorsale 1018 hPa sur l'ouest de l'Irlande et se renforce et s'oriente vers le sud prévue 1020 hPa en mer Celtique demain midi. Elle s'affaiblit sur le Golfe de Gascogne vendredi.	Rafales de vent jusqu'à 9 Beaufort au large de Gironde, orientées sud-ouest, s'atténuent progressivement vers le nord. Mer agitée devenant forte sur l'estuaire de la Loire cette nuit.
6	Dépression 1002 hPa au large du Portugal se déplace vers le nord-est prévue 998 hPa près du cap Finisterre ce soir. Elle se creuse davantage sur le golfe de Gascogne samedi matin.	Houle de nord-ouest 4 mètres au sud de la Sardaigne se propage vers l'est, atteignant 3,5 mètres près de la Sicile demain midi. Elle diminue sur la mer Ionienne en soirée.
7	Front froid 1012 hPa sur la mer d'Iroise progresse vers le sud-est prévu 1008 hPa sur les côtes vendéennes cette nuit. Il s'atténue sur le centre-ouest dimanche.	Coup de vent 7 à 8 Beaufort au large de la Minorque, prévu pour s'orienter nord-ouest avec des rafales à 10. Visibilité réduite sous averses sur Baléares ce soir.
8	Minimum 1005 hPa en mer Ligue se renforce et reste quasi-stationnaire prévu 1001 hPa au sud de Nice demain après-midi. Il se décale vers la Corse lundi soir.	Dépression « Bruno » 1000 hPa sur la mer Tyrrhénienne se déplace vers le sud-est, prévue 997 hPa au large de la Calabre jeudi matin. Elle fusionne avec un front sur la mer Adriatique vendredi.
9	Anticyclone 1028 hPa centré sur les Açores s'étend vers l'Europe prévue 1024 hPa sur le proche Atlantique ce week-end. Il se maintient sur l'ouest mardi prochain.	Mer très forte avec vagues de 5 mètres au nord de la Corse, causée par un vent de nord-est force 8. Conditions s'améliorant progressivement sur la Ligurie demain après-midi.
10	Dépression 1000 hPa au nord de l'Ecosse se déplace rapidement vers le sud-est prévue 995 hPa en mer du Nord demain matin. Elle s'évacue vers la Baltique mercredi.	Anticyclone subtropical 1025 hPa au large de Madère se décale vers le nord, prévu 1028 hPa sur les Canaries ce week-end. Temps stable et vent faible sur l'Atlantique est dimanche.

Figure 6: Results of Claude and Grok

5.2.1 Claude-opus-4: Canonical structure and genre fidelity

When prompted, Claude generates meteorological statements that, at first glance, closely emulate the rhetorical and phraseological conventions of French shipping forecasts. Several sentences display strong conformity to the genre's canonical sequencing: identification of the phenomenon, indication of central

pressure, localization, process predicate, and projected evolution, consistent with the structural regularities identified in corpus-driven studies of the field.

Sentence 1 (*Dépression 1009 hPa au nord de la Bretagne se renforce et se déplace vers l'est* […] – "Low-pressure system at 1009 hPa north of Brittany is strengthening and moving eastward […]") mirrors authentic formulations in which low-pressure systems are tracked through pressure evolution and displacement. The subsequent part (*Elle s'évacue vers la Mer du Nord* […] – "It evacuates toward the North Sea") reproduces the standardized expressions found in the corpus. In Sentence 3, Claude produces a genre-faithful representation of a low (*Dépression 1004 hPa en Méditerranée occidentale se creuse rapidement* […] – "Low-pressure system at 1004 hPa in the western Mediterranean is deepening rapidly"), echoing a pattern observed in the corpus, where Mediterranean or Genoa lows *se creusent* ("deepen") and *remontent* ("move") toward the Gulf of Genoa. Likewise, the treatment of *front froid* ("cold fronts") in Sentence 7 and *minimum dépréssionnaire* ("minima lows") in Sentence 8 demonstrated Claude's ability to mirror the genre's specialized terminology and fixed lexico-grammatical patterns.

However, Claude's apparent mastery of rhetorical templates masks significant conceptual inaccuracies arising from the absence of an underlying meteorological ontology. In Sentence 4 (*Thalweg 1010 hPa […] s'approfondit et se décale vers le nord-est […]* – "Trough at 1010 hPa […] is deepening and shifting northeastward"), Claude assigns a pressure value to a *thalweg* (trough), even though a *thalweg* does not denote a pressure center but rather an elongated axis of low pressure. As shown in the corpus, troughs are described in terms of their orientation and extension, not their central pressure. A similar issue emerges in Sentence 5, where Claude describes a *dorsale* ("ridge") as *s'orientant vers le sud* ("shifting southward"), a structure that is grammatically plausible yet uncommon in operational forecasts. Typically, ridges are described as *s'étendant* ("extending") or *s'allongeant* ("lengthening"), not as *orientés* ("oriented") like mobile entities. These deviations exemplify the type of frame mismatch described in frame semantics (Fillmore/Baker 2012; Fillmore 1976), in which lexical units evoke incompatible sets of FEs. As a result, the model fills in semantic roles (process, measure, movement) that the term's "true" semantic type does not warrant. Claude introduces non-attested pressure trajectories,

as in Sentence 2 and Sentence 9, where anticyclones (high-pressure systems) are assigned fluctuating pressures that are inconsistent with the slow temporal evolution observed in the corpus. This underscores the observation that the model reproduces surface templates without integrating domain-specific knowledge frames.

Although Claude mimics the rhetorical and phraseological structure of shipping forecasts, it fails to take account of the conceptual distinctions between key meteorological entities. In line with Frame-Based Terminology (L'Homme 2020; Faber 2015), these errors reveal that specialized terms cannot be interchanged without compromising semantic coherence of their argument structure. The model often breaks the mapping between linguistic form and expert knowledge, as observed in specialized genres (Orna-Montesinos 2012) This highlights the limits of template-based generation when semantic constraints are absent.

5.2.2 Grok-4-fast: Semantic fluency and discursive expansion

When prompted, Grok shows a mixed ability to replicate the rhetorical and terminological conventions of shipping forecasts. Several outputs follow the canonical structure of the genre. For instance, Sentence 1 (*Tempête 'ALEX' 995 hPa au large de Brest se déplace vers l'est* – "Storm 'ALEX' at 995 hPa off Brest is moving east") mirrors authentic patterns in which storms are tracked by pressure and movement. Sentence 2 (*Front froid 1012 hPa au nord de la Biscaye s'intensifie* […] – "Cold front at 1012 hPa north of the Bay of Biscay is intensifying") aligns with standard templates for describing frontal systems (Condamines 2018). Sentence 4, which depicts an anticyclone extending northeastward, also reflects phrasing observed in the corpus, using process and location.

However, several sentences confirm Grok's lack of meteorological ontology. In Sentence 6, it incorrectly attributes a movement predicate to *houle* ("swell"), which, in expert forecasts, denotes a state rather than an active agent, another example of frame mismatch in the sense of frame semantics (Fillmore 1976). Similarly, Sentence 3 deviates from domain-specific conceptual patterns by depicting a Genoa low deepening westward, a trajectory inconsistent with observed meteorological behavior. In Sentence 8, Grok states that a *dépression* ("low-pressure system") *fusionne* ("merges") with a front, an event unattested

in the corpus and semantically incompatible with Frame-Based Terminology, which holds that specialized units encode non-interchangeable conceptual frames (Faber 2015; L'Homme 2020). These deviations confirm that Grok reproduces phraseological structures without integrating expert knowledge.

Grok reproduces the formal and phraseological patterns of marine forecasts, but it lacks the epistemic precision needed to align lexical choices with the underlying conceptual frames of meteorology. As a result, it generates sequences that are syntactically plausible yet conceptually incorrect, revealing the gap between surface genre conformity and true terminological adequacy. In controlled-language contexts, such errors may have significant safety implications.

5.2.3 Comparison between Claude and Grok

To summarize the observations, this section compares the types of errors produced by Claude and Grok. While both models successfully reproduce the surface phraseology characteristic of the genre, their outputs diverge substantially in terms of conceptual accuracy, frame compatibility, and adherence to domain-specific constraints. The following comparison highlights the limitations of each system.

Error Type	Claude-opus-4	Grok-4-Fast
Frame Mismatch	Frequent: assigns pressure values to *thalweg*, uses dynamic predicates for non-dynamic entities (e.g. *dorsale s'oriente;* ridge shifts)	Frequent: treats *houle (swell)* as an agent, describes impossible trajectories (e.g. Genoa low deepening westward)
Conceptual Error	Moderate: introduces non-attested pressure variations; confuses semantic roles associated with meteorological categories	High: invents implausible events such as a *dépression* merging with a front; overpopulates semantic roles
Register Error	Occasional: uses generic geographical terms instead of nautical zones (e.g., *Méditerranée occidentale*)	Rare: mostly stays within genre-appropriate lexicon but may drift into verbose or overly narrative phrasing
General Trend	Strong control of terminology and structure but struggles with ontological consistency	Strong rhetorical fluency but conceptually less reliable

Figure 7: Error types produced by Claude-opus-4 and Grok-4-fast

Collectively, these errors point to different weaknesses in Claude and Grok: Each model exhibits a distinct balance between structural fidelity and conceptual robustness. This supports the idea that apparent genre mastery does not necessarily entail epistemic adequacy. These findings pave the way for the concluding discussion, where we reflect on the broader implications of these limitations and outline future directions for improving LLM performance in highly constrained, ontology-dependent domains.

6 Conclusion, limits and outlook

This study examined the capacity of LLMs to reproduce the lexico-grammatical regularities and discursive organization of French shipping forecasts under zero-shot conditions. Although both models demonstrated an impressive ability to replicate the surface mold of forecast discourse, including its structural segmentation, standardized phraseology, and syntactic compression, they also exhibited systematic conceptual limitations. Claude adhered more closely to terminological constraints but struggled with frame compatibility, whereas Grok produced rhetorically more fluent outputs yet displayed a tendency to misassign semantic roles and overpopulate predicates. In both cases, the lack of an internalized meteorological ontology resulted in conceptually inconsistent or implausible outputs despite surface fluency.

Several limitations temper the generalizability of these findings. First, the number of outputs generated was limited, and the analysis focused on only two model architectures; a broader sample and few-shot settings might increase the performance of the models. Second, the evaluation was conducted without fine-tuning or explicit semantic constraints, meaning that model performance reflects baseline general-purpose behavior rather than domain-specific optimization. Third, although grounded in empirical corpus data, the evaluation remains interpretative and dependent on linguistic, terminological, and conceptual expertise.

Building on these results, future research may examine semantic frame-based prompting as a means of guiding model generation with structured

knowledge representations. Integrating FrameNet-style annotation into prompts could help align lexical items with appropriate conceptual roles, reducing frame mismatches and enhancing epistemic validity. A final direction involves training smaller, domain-adapted models fine-tuned on annotated meteorological corpora, which may outperform general-purpose LLMs in specialized generation tasks while offering greater interpretability and operational control.

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